Ai Based Synthetic Media Detection System LABORATORY

DIGITAL MEDIA AUTHENTICATION & DEEPFAKE INVESTIGATION DOSSIER

Case Reference:	SMDS-DF-00018	**Date of Analysis:**	2026-09-23 09:06:24
Investigator/User:	Shri asar	**Media Type:**	IMAGE
Original Filename:	0250.png	**Face Detected:**	Yes (Primary ROI isolated)
SHA-256 Hash:	`d75e46b9c87963683a2cfa9a3683de51f0d19ee0edc7d87be66249a456943df9`		

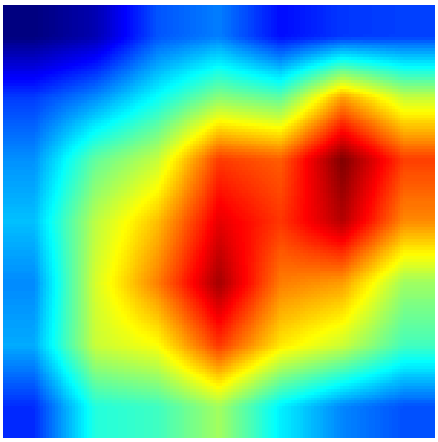
FORENSIC VERDICT: DEEPFAKE DETECTED

Confidence Level: **99.6%** | Fake Probability: **99.6%** | Real Probability: **0.4%**

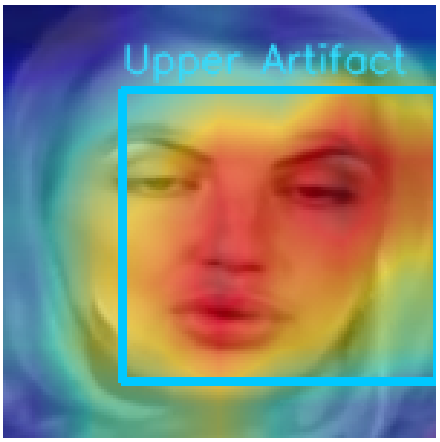
Visual Evidence & Explainable AI (Grad-CAM) Activation Maps



Isolated Facial ROI



Grad-CAM Heatmap



Suspicious Region Overlay

Forensic Anomaly & Artifact Distribution Breakdown

Forensic Indicator	Score / Value	Benchmark / Anomaly Criteria
Boundary Artifact Discontinuity	65.6%	> 35% indicates blending seams
High-Frequency Texture Variance	6.0%	Disparity from natural sensor noise profile
2D FFT Spectral Frequency Ratio	32.5%	Power spectrum peaks characteristic of GAN/Diffusion
Facial Bilateral Symmetry Coherence	33.3%	Structural landmark alignment discrepancy
Composite Deepfake Threat Index	**35.0%**	Fused spatial, spectral & boundary multi-metric risk
Risk Index Classification	**CRITICAL**	Computed aggregate forensic threat classification

Investigative Notes & Legal Advisory

This examination was performed using a calibrated deep convolutional neural network (ResNet-18 architecture) trained on manipulated facial sequences and validated against synthetic artifact distributions. The class activation maps highlight pixel regions contributing predominantly to the model classification. This certificate provides probabilistic evidentiary support for cyber-investigations and dispute resolution.

Status: Digitally Certified

Verification Stamp: SMDS-OK